

River Test

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Riverboat(60 points)

1. (6 points) A ferry is crossing a river that is 15m wide at a velocity of 3 ms^{-2} right relative to the water. However, the water is travelling at 3 ms^{-2} up relative to the ground. Calculate:
 - (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river
 - (c) The displacement of the boat
 - (d) The vertical displacement of the boat
2. (6 points) A ferry is crossing a river that is 192m wide at a velocity of 24 ms^{-2} right relative to the water. However, the water is travelling at 2 ms^{-2} up relative to the ground. Calculate:
 - (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river
 - (c) The displacement of the boat
 - (d) The vertical displacement of the boat
3. (6 points) A ferry is crossing a river that is 64m wide at a velocity of 16 ms^{-2} right relative to the water. However, the water is travelling at 8 ms^{-2} up relative to the ground. Calculate:
 - (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river
 - (c) The displacement of the boat
 - (d) The vertical displacement of the boat
4. (6 points) A ferry is crossing a river that is 24m wide at a velocity of 24 ms^{-2} right relative to the water. However, the water is travelling at 2 ms^{-2} up relative to the ground. Calculate:
 - (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river
 - (c) The displacement of the boat
 - (d) The vertical displacement of the boat
5. (6 points) A ferry is crossing a river that is 180m wide at a velocity of 18 ms^{-2} right relative to the water. However, the water is travelling at 7 ms^{-2} up relative to the ground. Calculate:
 - (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river
 - (c) The displacement of the boat
 - (d) The vertical displacement of the boat
6. (6 points) A ferry is crossing a river that is 100m wide at a velocity of 20 ms^{-2} right relative to the water. However, the water is travelling at 1 ms^{-2} up relative to the ground. Calculate:
 - (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river

- (c) The displacement of the boat
 - (d) The vertical displacement of the boat
7. (6 points) A ferry is crossing a river that is 24m wide at a velocity of 4 ms^{-2} right relative to the water. However, the water is travelling at 4 ms^{-2} up relative to the ground. Calculate:
- (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river
 - (c) The displacement of the boat
 - (d) The vertical displacement of the boat
8. (6 points) A ferry is crossing a river that is 1m wide at a velocity of 1 ms^{-2} right relative to the water. However, the water is travelling at 1 ms^{-2} up relative to the ground. Calculate:
- (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river
 - (c) The displacement of the boat
 - (d) The vertical displacement of the boat
9. (6 points) A ferry is crossing a river that is 7m wide at a velocity of 1 ms^{-2} right relative to the water. However, the water is travelling at 9 ms^{-2} up relative to the ground. Calculate:
- (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river
 - (c) The displacement of the boat
 - (d) The vertical displacement of the boat
10. (6 points) A ferry is crossing a river that is 18m wide at a velocity of 18 ms^{-2} right relative to the water. However, the water is travelling at 4 ms^{-2} up relative to the ground. Calculate:
- (a) The resultant velocity of the ferry relative to the ground
 - (b) The time taken to cross the river
 - (c) The displacement of the boat
 - (d) The vertical displacement of the boat